

Slide 1: Title & Societal Context

Title: From Face Perception to Social Categorization: A Multidimensional Scaling Approach to Investigate Ethnic Outgroup Homogeneity in Childhood

Authors & Affiliation

Societal Relevance:

- Increasing immigration and multicultural societies
- Early childhood: critical period for social categorization
- Understanding bias formation at the perceptual level

Slide 2: The Roots of Categorization

Social Categorization & Bias:

- Categorization has profound social consequences (Judd et al., 1988)
- Early-emerging ingroup preferences documented in infants (e.g., Kelly et al., 2009; Woo et al., 2020)

The Perceptual-Social Linkage:

- Infants' visual experience ↔ Later social biases (Lee et al., 2017)
- Visual specialization for ingroup faces (Quinn et al., 2016)
- Foundation for downstream intergroup attitudes (Timeo et al., 2017)

The Outgroup Homogeneity Effect (OHE; Quattrone & Jones, 1981)

- Outgroup faces appear more homogeneous to each other than ingroup faces
- Well-established in adults
- Mostly studied in dual Ingroup-Outgroup paradigms

The Outgroup Homogeneity Effect (with Fantastic Mr.Fox)

In-group



Outgroup Beans



Outgroup Boggins



Outgroup Bunce



Minority-groups homogeneity effect (Tepper & Gilovich, 2025)

- In a multi-group scenario
- Majority members see different minority-outgroups as MORE similar among themselves

Minority-groups homogeneity effect (with Fantastic Mr.Fox)



The Multigroup Context

In children, the situation is slightly different:

- No formal position consolidated
- Multi-group paradigm is not employed, even if it is more ecological
- **Gap:** We don't know how children categorize faces in a multi-group environment

How to study this effect in a multigroup system with a children population?

Study Rationale & Framework

Theoretical Shift: From Norm-Based to Exemplar Models

- Critique: Forced-choice, dichotomous paradigms miss relational complexity
- Face-Space Theory (Valentine et al., 2016): Faces exist in a continuous multidimensional perceptual space
- Exemplar-based model

Research Question:

- How is the multiethnic perceptual space structured in childhood?
- Does the ingroup occupy a unique position in children's face-space?

Slide 5: Key Methodological Innovations

Multidimensional Scaling (MDS) for Developmental Research:

- First application to trace social categorization trajectories in early childhood
- Maps three-dimensional spatial relationships of perceived similarity

Expanding the Scope:

- 4-group context: Italian ingroup + 3 distinct ethnic outgroups
- Relational focus: Facial dyads rather than isolated face evaluation
- Continuous similarity judgments (0-3 scale) instead of binary choices

Pre-registered studies: Method & Experimental Design

Sample:

- Study 1: N=249 (moderated, controlled lab setting)
- Study 2: N=101 (unmoderated, online validation)
- Age range: 3 to 8 years

Stimuli: AI-Generated Faces

- 40 faces per group (10 per group, controlled morphologies)
- Groups: Italian (ingroup), African Descent, East Asian, Andean
- FaceNet verification: Ensured perceptual distinctiveness

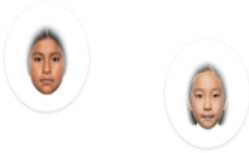
Validity & Replicability:

The Gamified Task

Engagement Without Explicit Categorization:

- Narrative: "Oscar the meerkat" task (no explicit ethnic labels)
- Children evaluate face pairs without social priming

The Face Similarity Task

→	→	→
+		Quanto si somigliano? Molto DiversiUn po' DiversiUn po' SimiliMolto Simili

Face Similarity Task Protocol:

- 66 random dyads presented sequentially
- Response scale: 0–3 (simple visual scale for ages 3+)
- ~10 minutes per session

MDS

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<iframe src="mds-interactive.html" title="Interactive 3D MDS visualization"  
width="100%" height="560" style="border: 0;" > </iframe>
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Slide 8: Results - 3D Spatial Mapping (H1A)

Key Finding: OHE NOT Supported

- Italian ingroup: Highly concentrated cluster
- African Descent outgroup: Highly concentrated cluster
- Andean outgroup: **Largest dispersion volume** (lowest homogeneity)
- East Asian outgroup: Intermediate compactness

Interpretation:

- Perceptual homogeneity $\neq$ predicted by ingroup/outgroup status
- Suggests facial morphology and feature distinctiveness matter more than social category

Slide 9: Results - Group Distances (H2A)

Key Finding: MGHE NOT Supported

- Outgroups were NOT closer to each other than to the ingroup
- African Descent group: Structurally most distant from other outgroups
- Ingroup-Outgroup distances: Heterogeneous (not uniformly large)

What This Means:

- Minority groups not perceived as interchangeably similar
- Each group occupies distinct perceptual position
- Multigroup structure more complex than binary ingroup/outgroup dichotomy

Slide 10: Results - Developmental Trajectories

Linear Mixed Models: Developmental Changes

Intergroup Dyads (In-Out, Out-Out):

- Significantly MORE dissimilar as age increases
- Suggests sharpening of perceptual distinctions

Intragroup Dyads (In-In, Out-Out):

- Study 1: Remain stable across ages
- Study 2: In-In distance decreases with age
- May indicate consolidation of category boundaries

Interpretation:

- Perceptual expertise increases with age

Slide 11: Structural Replicability (S1 vs S2)

Validating Robustness Across Contexts

Procrustes 3D MDS Alignment:

- Highly consistent structural overlap between moderated (S1) and unmoderated (S2) samples
- Similar spatial arrangements despite different testing contexts

Quantitative Validation:

- Tucker congruence indices: >0.93 (excellent agreement)
- Mantel tests: Confirm consistent mapping structures
- **Conclusion:** Perceptual structures are robust, not artifacts of testing context

Slide 12: Discussion & Conclusion

Main Findings:

- Visual essentialism & face-space similarity better explain results than OHE/MGHE
- Multigroup perceptual space: Complex, group-specific organization
- Developmental refinement: Intergroup distinctions sharpen with age

Limitations:

- Uneven dyad exposures in similarity task
- Perceived threat/cultural factors not directly measured

Future Directions:

- Interventions in multicultural schools based on perceptual similarity principles
- Integration with implicit attitude measures

Reference Document: Copia di ENG_Scaletta_Art1_CH_27_08_26 (3).docx